Generic mixed-degree field preservation

The actual weighted K-nonzero source formula now preserves normalized Mumford validity from original valid inputs. Nonzero output weights and guarded Some return are also proved. No output identities or validity are assumed.

Construction. Put $u = X^2 + qX + r$, $v = sX + t$, $k = u(x) \neq 0$, $m = (y - v(x))/k$ and $h = v + mu$. An explicit Bezout identity derives $(X - x)u \mid f - h^2$. Define

$$q' = x - q - m^2, \quad r' = a_3 + q^2 - r - m(2s + mq) + xq'.$$

The residual $f - h^2 - (X - x)u(X^2 + q'X + r')$ has degree at most2 and is divisible by a degree3 polynomial, so vanishes. Reducing $-h$ gives $s' = mq' - (s + mq)$ and $t' = mr' - (t + mr)$; factorization proves output validity.

Weighted source. genericK preserves the generic source association and is proved equal to the cancellation prelude guard. All four output normalization identities follow from the literal weighted formula. The normalized slope is rawA/outputZ. Nonzero rawK and weights imply nonzero outputZ and outputW=1. The final theorem composes these facts with original weighted input divisibilities against a monic quintic without an X4 term.

Controls. Nonunit a=(0,64,2,2), b=(0,81,0,0,3,2), f=$X^5 + X^2 - X + 1$ normalize to point(0,1), support$X^2 + 1$, ordinate0. Explicit input witnesses instantiate the full theorem; output normalizes to(-1,-1,-1,-2). A zero-slope full affine control yields repeated output support. ZeroK returnsNone; an altered constant coefficient fails point validity.

Receipts. Final module, controls and separate readback exit0; only propext, Classical.choice and Quot.sound. An erroneous scratch fixture expansion was corrected in a separate frozen control plan and kernel-checked. Namespace/probe/local-definition failures remain recorded. All compiles bounded120seconds, existing pinned dependencies.

Scope. Field polynomial validity, without smoothness or divisor-class claim. Literal UInt64, compiler/counter, tripling and whole-walk obligations remain separate. Independent review pending. No scientific trials or costs. Exact source gate232–292 and primary paper references are in sources.json/PLAN.md; theorem ledger in status.json.